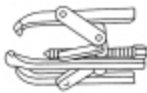
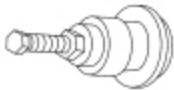


Crankshaft Pulley



Special Tool(s)

 ST1184-A	3 Jaw Puller 303-D121 or equivalent
 ST1328-A	Installer, Front Cover Oil Seal 303-335 (T88T-6701-A2 plate only)
 ST3249-A	Installer, Front Crank Seal and Damper 303-1531
 ST1438-A	Strap Wrench 303-D055 (D85L-6000-A) or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

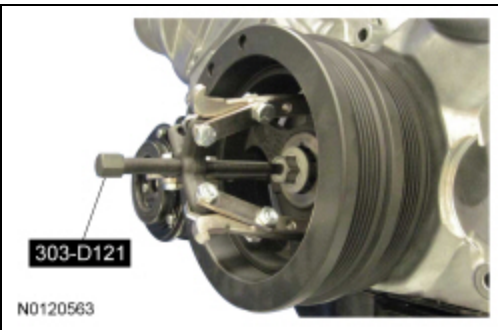
Material

Item	Specification
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS- M2C945-A
Motorcraft® Silicone Gasket Remover ZC-30-A	—
Motorcraft® Silicone Gasket and Sealant TA-30	WSE- M4G323-A4

Removal

NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces, that enters the oil passages, coolant passages or the oil pan, can cause engine failure.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. Remove the accessory drive and A/C compressor belts. For additional information, refer to Section 303-05.
3. Remove the crankshaft pulley bolt and washer.
4. Using the 3 Jaw Puller and the crankshaft pulley bolt, remove the crankshaft pulley.
 - Discard the crankshaft pulley bolt.



Installation

1. **NOTE:** If not secured within 5 minutes, the sealant must be removed and the sealing area cleaned with silicone gasket remover and metal surface prep. Failure to follow this procedure can cause future oil leakage.

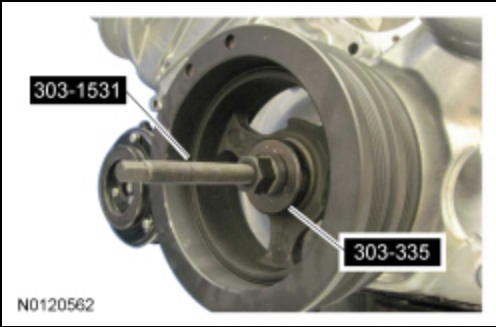
Apply silicone sealer to the Woodruff key slot in the crankshaft pulley.



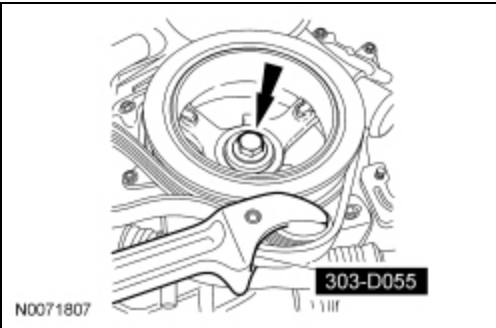
2. Lubricate the crankshaft pulley sealing surface with clean engine oil prior to installation.



3. Using the Front Crank Seal and Damper Installer and the Front Cover Oil Seal Installer (plate only), install the crankshaft pulley.



4. Using the Strap Wrench, install a new crankshaft pulley bolt and the original washer, tighten the bolt in 4 stages:
- Stage 1: Tighten to 140 Nm (103 lb-ft).
 - Stage 2: Loosen 360 degrees.
 - Stage 3: Tighten to 100 Nm (74 lb-ft).
 - Stage 4: Tighten an additional 90 degrees.



5. Install the accessory drive and A/C compressor belts. For additional information, refer to Section 303-05.
6. After completing the repairs, use the scan tool to perform the Misfire Monitor Neutral Profile Correction procedure, following the on-screen instructions.